

1 Heegaard Diagrams of 3-Manifolds

Example 1 ($L(\mathbf{p}, \mathbf{q}) := \mathbf{S}^3 / [(\mathbf{w}, \mathbf{z}) \sim (\xi_{\mathbf{p}} \mathbf{w}, \xi_{\mathbf{q}} \mathbf{z})]$). Consider the standard Heegaard decomposition of S^3 given by

$$S^3 = \underbrace{\{|z| \geq |w|\}}_{h^1} \cup \underbrace{|z| = |w|}_{\mathbb{T}^2} \cup \underbrace{\{|w| \geq |z|\}}_{h^2} \quad (1)$$

Then the action of $\mathbb{Z}/(p)$ which I will denote by f preserves the handlebodies in this decomposition alongside the torus $|z| = |w|$. This gives us

$$L(p, q) = h^1 / f \cup_{\mathbb{T}^2 / f} h^2 / f \quad (2)$$

This is useless unless we can describe what these sets look like, notice that if we can describe $\mathbb{T}^2 / f$, then we have also described the handlebodies, since these can be expressed as $\mathbb{T}^2 / f \times I$. Computing for $\mathbb{T}^2$:

$$\mathbb{T}^2 / f = \frac{\mathbb{T}^2}{(x, y) \sim \left(x + \frac{1}{p}, y + \frac{q}{p}\right)} = \frac{\mathbb{R}^2 / (\mathbb{Z}^2)}{\left(\frac{1}{p}, \frac{q}{p}\right)\mathbb{Z}} = \frac{\mathbb{R}^2}{(0, 1)\mathbb{Z} \oplus \left(\frac{1}{p}, \frac{q}{p}\right)\mathbb{Z}} \cong \mathbb{T}^2 \quad (3)$$

Thus equation (2) also specifies a Heegaard Decomposition. Using the basis computed in equation (3) we take the corresponding generators for $H_1(\partial h^j)$. Now $(0, 1)\mathbb{Z}$ is the meridian for h^1 and thus also h^1 / f , same for $(1, 0)$ as the meridian of h^2 , h^2 / f . The meridian of h^2 is thus glued to $p(1/p, q/p) - q(0, 1)$. This gives the heegaard decomposition on $\mathbb{T}^2$ as the curves given by $(\alpha, p\beta - q\alpha)$, where $\alpha, \beta \in H^1(\mathbb{T}^2)$ are the homology basis of the quotient computed in equation (3).

Example 2 ($\mathbb{T}^3$ with the flat metric). Model $\mathbb{T}^3$ as $[0, 2\pi]^3$, then consider

$$f(x) = \sum_1^3 \frac{1}{2} \cos(x^j) + \frac{3}{2} \quad (4)$$

I claim that this is a self-indexing morse function, checking this

$$-\nabla f = \frac{1}{2} \sin(x^j) dx_j \implies (-\nabla f)^{-1}(0) = \{(s, t, r) \mid s, t, r \in \{0, \pi\}\} \quad (5)$$

And at such points we have

$$\text{Hess}(f) = \frac{1}{2} \text{Diag}(-\cos(x^j)) \quad (6)$$

This verifies f is morse, to see its self indexing note that the index is the number of occurrences of 0 in the critical point vector.

Now we solve for a gradient flow of f , from the above computations

$$\dot{x}^j = \frac{1}{2} \sin(x^j) \quad (7)$$

We can apply separation of variables to $y' / \sin y = \frac{1}{2}$ this gives us $\int \frac{1}{\sin(x^j)} dx^j = \frac{t}{2} + C$ integrating gives us

$$\log(\tan(x^j/2)) = \frac{t}{2} + C \quad (8)$$

and thus

$$x^j = 2 \arctan(Ce^{t/2}) \quad (9)$$

Using the initial condition $x^j(0), C = \tan(x^j(0)/2)$ so that

$$x^j = 2 \arctan(\tan(x^j(0)/2)e^{t/2}) \quad (10)$$

The Heegaard diagram uses the asymptotic information of gradient flows, luckily this is easy to compute:

$$\lim_{t \rightarrow \infty} x^j = \begin{cases} \pi & x^j(0) \neq 0 \\ 0 & x^j(0) = 0 \end{cases} \quad \lim_{t \rightarrow -\infty} x^j = \begin{cases} 0 & x^j(0) \neq \pi \\ \pi & x^j(0) = \pi \end{cases} \quad (11)$$

(We make sense of this by thinking of S^1 as the codomain of $\tan$ and domain of $\arctan$) It follows that the stable manifold of the index one critical point $(0, \pi, \pi)$ is given by $\{(0, x^2, x^3) \mid x^2, x^3 \neq 0\}$. The others are described symmetrically. Similarly we can consider the unstable manifold of the critical point $(\pi, 0, 0)$ as $\{(\pi, x^2, x^3) \mid x^2, x^3 \neq \pi\}$, the rest of the cases are symmetric.

Now, to get the Heegaard decomposition, we study the level set $f^{-1}(3/2)$, the α curves are given by

$$\alpha_1 = \{(0, x^2, x^3) \mid \cos(x^2) + \cos(x^3) = -1\} \quad (12)$$

$$\alpha_2 = \{(x^1, 0, x^3) \mid \cos(x^1) + \cos(x^3) = -1\} \quad (13)$$

$$\alpha_3 = \{(x^1, x^2, 0) \mid \cos(x^1) + \cos(x^2) = -1\} \quad (14)$$

And the β curves

$$\beta_1 = \{(\pi, x^2, x^3) \mid \cos(x^2) + \cos(x^3) = 1\} \quad (15)$$

$$\beta_2 = \{(x^1, \pi, x^3) \mid \cos(x^1) + \cos(x^3) = 1\} \quad (16)$$

$$\beta_3 = \{(x^1, x^2, \pi) \mid \cos(x^1) + \cos(x^2) = 1\} \quad (17)$$